Tool Gap Analysis: Automotive Physical AI

What transfers directly, what needs adaptation, and what is missing from the automotive tool stack

Transfers Directly

~70% of methodology

V-Model Lifecycle — TRANSFERS

- Process phases (concept → decommission)
- Work product structure
- Confirmation measures

Risk Analysis Methods — TRANSFERS

- HARA methodology → Risk Assessment
- FMEA/FMEDA/FTA analysis
- Safety goal derivation

SW Development Process — TRANSFERS

- MISRA C/C++ coding standards
- Static/dynamic analysis
- Unit/integration testing

Tool Qualification — TRANSFERS

- TCL determination (ISO 26262-8 Cl.11)
- Tool confidence levels
- Qualification evidence packages

Cybersecurity (21434) — TRANSFERS

- TARA methodology
- Security requirements
- Vulnerability management

AI/ML Safety (8800) — TRANSFERS

- Data quality requirements
- Model verification
- Performance monitoring

Needs Adaptation

~20% requires rework

Hardware Metrics — ADAPT

- SPFM/LFM/PMHF → PFHd via MTTFd
- Diagnostic coverage mapping
- Category architecture (B,1-4)

Integrity Derivation — ADAPT

- HARA (S×E×C) → Risk Graph (S×F×P)
- ASIL → PL mapping
- Different controllability models

Validation Environment — ADAPT

- Proving ground → Physical workspace
- Driving scenarios → Robot tasks
- ODD → Collaborative workspace

Architecture Patterns — ADAPT

- ASIL decomposition → Cat. 3/4
- Dual-channel vs. monitored
- Safe state differences

Missing from Automotive

~10% entirely new

Safety PLCs & Logic Solvers — NEW

- Siemens F-CPU (S7-1500F)
- Allen-Bradley GuardLogix
- Pilz PSS 4000

SISTEMA Calculation — NEW

- PL verification software (IFA)
- Category/DC/CCF/MTTFd calc
- Block diagram modeling

Workspace Monitoring — NEW

- 3D safety zone monitoring
- Light curtains / laser scanners
- Pressure-sensing mats

Force/Torque Limiting — NEW

- ISO/TS 15066 force thresholds
- Contact force measurement
- Power & force limiting (PFL)

Industrial Protocols — NEW

- EtherCAT / PROFINET / PROFISAFE
- OPC UA / OPC UA Safety
- IO-Link Safety

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